Addison Vang

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EDUCATION

University of Wisconsin-Stout **Expected: May 2027**
B.S. in Applied Mathematics and Computer Science

- Relevant Coursework: Data Structures, Linear Algebra, Numerical Analysis, Machine Learning, Operating Systems, Computer Architecture
- Extracurricular Activities: President of the Hmong Stout Student Organization (2024 - 2026)

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript, HTML/CSS, PHP, SQL, R
Tools: Visual Studio Code, Jupyter Notebook, Google Colab, MySQL, Linux

EXPERIENCE

University of Wisconsin-Stout **Feb 2024 – Present**
Math Tutor Menomonie, WI

- Provided one-on-one math tutoring through Student Support Services, covering calculus through linear algebra; developed clear communication skills working with diverse students.

3M Assembly **Jun 2025 – Aug 2025**
3rd Shift Assembly Red Wing, MN

- Maintained strong attention to detail in a fast-paced production environment, following safety protocols and adapting to rotating tasks and production demands.

University of Wisconsin-Eau Claire **Jun 2024 – Jul 2024**
Tutor/Mentor Eau Claire, WI

- Tutored and mentored high school students; led academic topic discussions and planned enrichment activities for the program.

PROJECTS

Pokemon Battle Outcome Predictor
Technologies: Python, scikit-learn, NumPy, Pandas

- Parsed 42,883 Pokémon Showdown battle logs to extract team rosters and outcomes, engineering 150-dimensional difference vectors covering 88.4% of all Pokémon appearances.
- Trained and compared three classifiers (Logistic Regression, Decision Tree, Softmax Neural Network) on an 80/20 stratified split; Logistic Regression achieved best accuracy (~54%) and ROC-AUC.

JPEG Image Compression via DCT
Technologies: Python, NumPy, SciPy, Matplotlib

- Built a JPEG-style compression pipeline from scratch: grayscale conversion, 8×8 tile segmentation, 2D DCT with orthonormal normalization, low-pass filtering, and inverse DCT reconstruction.
- Visualized DCT coefficient blocks via histogram equalization, demonstrating applied knowledge of signal processing and linear algebra.

Desktop Calendar Widget
Technologies: Electron, JavaScript, HTML, CSS

- Built a frameless, transparent desktop overlay using Electron that dynamically displays the current date and weekday; configured a fixed 214×228px borderless, context-isolated window for a lightweight always-on widget.